

VICTOR KIPRUTO ROP

Data Engineer

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SUMMARY

Data Engineering intern at Openfloat Limited and final-year BSc Data Science student who shipped PesaGuard, a real-time M-Pesa reconciliation and anomaly-detection SaaS now running with a live pilot customer. Builds production-grade streaming and batch data platforms for East African fintech using Apache Kafka, PyFlink, Airflow, and Terraform-provisioned Kubernetes, alongside dimensional data warehouse design and large-scale batch ETL (1.6M+ rows). Publishes technical writing on AWS Builder Center and holds IBM certifications in Big Data Foundations and Enterprise Design Thinking. Comfortable owning a pipeline from ingestion through CI/CD-tested, monitored production deployment.

TECHNICAL SKILLS

- **Languages:** Python, SQL (PostgreSQL)
- **Streaming & Orchestration:** Apache Kafka, Apache Flink (PyFlink), Apache Airflow (DAG design), Celery + Redis
- **Cloud & Infrastructure:** AWS (S3, EC2), Terraform (IaC), Kubernetes, Kustomize, Docker, Docker Compose, GitHub Actions (CI/CD)
- **Data Warehousing & Modeling:** Star Schema, Dimensional Modelling, SCD Type 2, Fact & Dimension Tables, dbt
- **Application & APIs:** Django 5 + DRF, Flask, Safaricom Daraja / M-Pesa API (OAuth2 / token-based auth), webhook-driven event ingestion, JWT auth, REST/OpenAPI
- **Data Stores:** PostgreSQL, Supabase, AWS S3 buckets
- **Version Control & CI/CD:** Git, GitHub Actions, GitHub
- **Data Quality & Testing:** Idempotent upserts, deduplication, validation rule engines, Pytest (unit & integration)
- **Visualization:** Power BI, Matplotlib, Tableau, Jupyter Notebooks, Grafana

PROFESSIONAL EXPERIENCE

Data Engineering Intern — Openfloat Limited | Nairobi, Kenya | June 2026 – Present

- Build and maintain data pipelines handling transaction processing and streaming data as part of the Data Engineering team during an industrial attachment.
- Support pipeline reliability and data quality in a live production engineering environment.

PROJECTS

PesaGuard — Real-Time M-Pesa Reconciliation & Anomaly Detection SaaS | Python/Flask, Next.js 14, Safaricom Daraja API

- Built a reconciliation and anomaly-detection platform for M-Pesa transactions, integrating directly with the Safaricom Daraja API (OAuth2 token-based authentication) via webhook-driven, event-based transaction ingestion.
- Took the product from architecture through to a working MVP now running with a live pilot customer.
- Prioritized production hardening ahead of new features: idempotent webhook handling, input validation on webhook payloads, a traceable audit trail, and alerting verified to fire on real conditions.
- Open-sourced the underlying streaming engine, which has drawn 21 GitHub stars.

AWS Streaming Fraud Detection — Real-Time M-Pesa Transaction Streaming Pipeline | Apache Kafka, Apache Flink, Terraform, Kubernetes, dbt

- Built an event-driven pipeline ingesting M-Pesa transaction webhooks from the Safaricom Daraja API via a Flask receiver, streamed through Kafka producers/consumers, and processed in real time with Apache Flink.
- Modeled the analytics layer with dbt on PostgreSQL/BigQuery and visualized it in Grafana, with the full stack provisioned via Terraform onto Kubernetes.
- Open-sourced the pipeline (21 GitHub stars) with Dockerized services, a health-check endpoint, and a documented sample webhook flow.
- Authored a companion technical deep-dive, “Building a Real-Time M-Pesa Transaction Pipeline: Kafka, PyFlink, and Terraform on Kubernetes,” published on AWS Builder Center.

M-Pesa Safaricom Fraud & Anomaly Detection Pipeline | Python, Apache Airflow, PostgreSQL, Docker

- Built an end-to-end M-Pesa data pipeline (extract, transform, validate, load) orchestrated with Airflow DAGs into PostgreSQL.
- Developed a dedicated fraud and anomaly-detection module alongside companion modules for agent network analytics, float liquidity forecasting, merchant intelligence, and remittance analytics.

- Enforced data quality with duplicate detection, null validation, schema checks, and transaction-integrity rules; fully containerized with Docker Compose.

Cloud ETL Pipeline (AWS) — Large-Scale Data Processing | *Python, PostgreSQL, Docker, connection pooling*

- Processed 1.6M+ rows across 4 source files at ~70K rows/sec with only 300MB peak memory, using chunked reading and automatic encoding detection.
- Cut database load time from an estimated 4+ minutes to 45 seconds via connection pooling and 5,000-row batch inserts, with exponential-backoff retry logic for transient failures.
- Built a modular Extract/Transform/Load architecture with independent unit tests, statistics tracking per stage, and full Docker containerization.

Data Engineering Projects Portfolio — Star-Schema Warehousing & Orchestration | *Python, Apache Airflow, dbt, PostgreSQL, Docker, GitHub Actions*

- Built a 15-project data engineering portfolio spanning three learning tracks — foundational ETL, intermediate warehousing/orchestration, and production-hardened pipelines — earning 26 GitHub stars.
- Designed a star-schema data warehouse with dimensional modeling, DAG-based orchestration in Apache Airflow, resilient paginated API integrations, and time-series IoT ingestion.
- Standardized every project with pytest test suites and a shared GitHub Actions CI workflow, runnable locally or via Docker Compose.

TECHNICAL WRITING

- “Building a Real-Time M-Pesa Transaction Pipeline: Kafka, PyFlink, and Terraform on Kubernetes” — **AWS Builder Center**

EDUCATION

BSc Data Science, Cooperative University of Kenya | *Expected 2027*

Kenya Certificate of Secondary Education (KCSE), St. Patrick's High School, Iten | *Graduated 2023*

CERTIFICATIONS

Big Data Foundations | *IBM / Cognitive Class* | 2026

- Comprehensive training in big data concepts, the Hadoop ecosystem, and distributed computing fundamentals.

Enterprise Design Thinking | *IBM* | 2026

- User-centered design principles and collaboration techniques for solving complex business problems.

Professional Development — Data Engineering & Systems Architecture | *Verified Credential* | 2026

- Continued learning and specialization in scalable data systems and engineering best practices.

ACTIVITIES

- GitHub Student Developer Pack — active, verified through early 2028
- Data Science Specialisation — Python, data analysis, machine learning, statistics
- Member, Data Science Club, Cooperative University — data challenges, workshops, and industry guest lectures
- Volunteer Tutor, Community Tutoring Program, St. Patrick's High School — mathematics and science